Case Study: Running an Audit Method Through AI

CAPSTONE-01 | AI-Assisted Audit Control Framework · AI Data Lab: Data Analysis

Deliverable: two-layer risk-register dashboard + five method records · **grc_risk_register.csv** — synthetic, 160 rows × 19 columns · N. Fragoso

1 WHAT THIS PROJECT WAS

AI will produce a defensible-looking analysis from almost any file. None of that makes a result safe in front of an audit committee. This project tested whether the discipline I use in regulatory audit and exams — understand the population, fix the question in writing, run the work, have it challenged independently, build the artifact from what survives — holds when the analysis itself is executed by AI. The dataset is a synthetic GRC risk register generated with Claude so no employer or production data is involved. The output is one self-contained dashboard with an executive and an auditor layer, plus a method record for each of the five steps. The records are the point: they make the analysis reviewable by someone who was not in the room.

2 THE FIVE STEPS, AND WHY EACH ONE EXISTS

Explore, Frame, Analyze, Verify, Visualize. This is the sequence I run for any analysis that will be seen by people who can act on it. Each step closes a specific failure mode — analysing a population I do not understand, choosing the question after seeing the data, leaving my own work unchallenged, and drawing a picture that says more than the evidence does. No step starts until the one before it is written down.

1 Explore	**Understand the population before touching it.** I profile the file first: what a single row represents, which columns are genuinely observed and which are derived from other columns, where the business rules hold and where they break. This is what tells me which analyses are meaningless before I spend a week producing them. A file can pass every integrity check and still be unusable for the question I want to ask.
2 Frame	**Fix the question before the analysis, in writing.** Who the audience is, what decision sits in front of them, which metric answers it, and which population every figure is computed over. Writing that down first is the control. It stops the question from being quietly reshaped by whatever the data turns out to show, and it stops a denominator from being chosen because it produced a better number.
3 Analyze	**Run the work, and keep the record of what changed while running it.** Every figure is regenerated from source in a single run rather than transcribed by hand, so no two artifacts can disagree with each other or with the file. Scope changes, corrections, and claims I overstated and pulled back mid-build are documented as they happen. The point of a workpaper is that someone can follow the reasoning, not just the result.
4 Verify	**Hand it to an adversary that does not know why I did what I did.** I ran the challenge with a separate agent, given no access to my reasoning from the first three steps and briefed only to attack the conclusions. It worked in two passes: first against the artifact alone, testing whether each claim was actually supported by what was on the page, then against the source file, recomputing everything. A second line of defence is only worth running if the challenger is genuinely independent of the person who did the work.
5 Visualize	**Build only what survived, and design against the wrong conclusion.** The dashboard was built after the challenge, not before, so anything the review withdrew is simply absent rather than annotated. It carries two layers over one computation run — an executive layer that states the conclusions plainly and an auditor layer that proves them — because leadership and the committee need different densities of the same truth.

3 HOW THE AI WAS USED

CLAUDE (OPUS 5) DID

- Wrote and ran every profiling and analysis script in pandas, plus the statistical testing — chi-squared, Fisher exact, Kruskal-Wallis, Cochran-Mantel-Haenszel, Clopper-Pearson intervals, Bonferroni correction.
- Generated the dashboard and all five records natively in code, so no rendering step can alter a number.
- Ran the adversarial challenge as a separate agent with no access to the reasoning behind the first three steps, against a brief and a blindness rule I set.
- Drafted every document.

I SET, DIRECTED AND OWN

- The question, audience, scope boundary and denominator rule — fixed in writing before analysis, so findings answer a question defined in advance.
- The two-pass review structure and what each pass was permitted to see.
- What could be charted, what had to be withdrawn, what could headline, and the confidence stated on each record.
- Every finding, withdrawal and recorded error, signed on the face of each document.

4 WHAT THIS DOES NOT CLAIM

The dataset is synthetic and three of its fields are not independent. Testing showed Control_Effectiveness is coupled to status and to nothing else in the file — not inherent score, not residual reduction, not exposure. The pack demonstrates a method; it is not evidence about how controls behave in a real institution. That sits in the dashboard masthead rather than the footer, because it changes how every figure below should be read.

SOURCE AND ACCOUNTABILITY. Synthetic GRC risk register built for portfolio demonstration; no production or employer data. Analysis, adversarial review, scripts and drafting performed with Claude (Opus 5) under human direction, Pass 1 of the review blind to source. Framing decisions, findings, withdrawals and the recorded reviewer error reviewed and owned by N. Fragoso. The five method records and the dashboard are published alongside this case study.